

DECODELABS TASK 04 – SYSTEM VULNERABILITY CHECKLIST

CODE:

```
# Cyber Security Project 4 by Harini Kasi
```

```
# System Vulnerability Checklist
```

```
print("===== SYSTEM VULNERABILITY CHECKLIST =====")
```

```
vulnerabilities = []
```

```
# Password Check
```

```
password = input("\nEnter system password for analysis: ")
```

```
has_upper = False
```

```
has_digit = False
```

```
has_symbol = False
```

```
for char in password:
```

```
    if char.isupper():
```

```
        has_upper = True
```

```
    elif char.isdigit():
```

```
        has_digit = True
```

```
    elif char in "!@#$%^&*":
```

```
        has_symbol = True
```

```
if len(password) < 8:
```

```
    vulnerabilities.append(
```

```
        "Weak password: Password length is less than 8 characters"
```

```
    )
```

```
if not has_upper:
```

```
    vulnerabilities.append(  
        "Weak password: Missing uppercase letters"  
    )
```

```
if not has_digit:
```

```
    vulnerabilities.append(  
        "Weak password: Missing numbers"  
    )
```

```
if not has_symbol:
```

```
    vulnerabilities.append(  
        "Weak password: Missing special characters"  
    )
```

```
# Software Update Check
```

```
update = input(  
    "\nIs the system software updated? (yes/no): "  
)
```

```
if update.lower() == "no":
```

```
    vulnerabilities.append(  
        "Software is outdated. Update security patches."  
    )
```

```
# User Practice Check
```

```
print("\nAnswer Security Practice Questions")

password_share = input(
    "Do you share passwords with others? (yes/no): "
)

if password_share.lower() == "yes":

    vulnerabilities.append(
        "Unsafe practice: Password sharing detected"
    )

unknown_links = input(
    "Do you open unknown links/emails? (yes/no): "
)

if unknown_links.lower() == "yes":

    vulnerabilities.append(
        "Unsafe practice: Risk of phishing attack"
    )

public_wifi = input(
    "Do you use public WiFi without protection? (yes/no): "
)

if public_wifi.lower() == "yes":

    vulnerabilities.append(
        "Unsafe practice: Public WiFi security risk"
    )
```

```
# Final Report

print("\n\n===== SECURITY AUDIT REPORT =====")

if len(vulnerabilities) == 0:

    print("System Status: SAFE")
    print("No vulnerabilities detected.")

elif len(vulnerabilities) <= 3:

    print("System Status: MEDIUM RISK")

else:

    print("System Status: HIGH RISK")

print("\nDetected Issues:")

for issue in vulnerabilities:

    print("-", issue)

print("\nTotal Vulnerabilities Found:",
      len(vulnerabilities))

print(
    "\nRecommendation: Fix the above issues "

    "to improve system security."
)
```

OUTPUT

```
PS D:\DecodeLabs> python -u "d:\DecodeLabs\Task4_system_vulnerability_checklist.py"
===== SYSTEM VULNERABILITY CHECKLIST =====

Enter system password for analysis: harini21

Is the system software updated? (yes/no): no

Answer Security Practice Questions
Do you share passwords with others? (yes/no): yes
Do you open unknown links/emails? (yes/no): no
Do you use public WiFi without protection? (yes/no): yes

===== SECURITY AUDIT REPORT =====
System Status: HIGH RISK

Detected Issues:
- Weak password: Missing uppercase letters
- Weak password: Missing special characters
- Software is outdated. Update security patches.
- Unsafe practice: Password sharing detected
- Unsafe practice: Public WiFi security risk

Total Vulnerabilities Found: 5

Recommendation: Fix the above issues to improve system security.
PS D:\DecodeLabs> 
```